

YASH BAVADIYA

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EDUCATION

Indian Institute of Technology, Jodhpur

Aug 2024 – Jul 2028

Bachelor's Degree - Computer Science and Engineering - CGPA - 7.21

Jodhpur, Rajasthan

TECHNICAL SKILLS

Languages: C++, C, Python, JavaScript, TypeScript, SQL

Core CS: Data Structures and Algorithms, Object-Oriented Programming, Software Design, Operating Systems, Database Management Systems, Computer Networks, Distributed Systems

AI Engineering: Prompt Engineering, Retrieval-Augmented Generation, Agentic Pipelines, LLM Evaluation, Context Management, FAISS, Hugging Face, PyTorch

Technologies/Frameworks: Qt (C++), CMake, SDL2, GoogleTest, Node.js, Express, React, PostgreSQL, MySQL, MongoDB, Redis, WebSockets, Docker, Linux/UNIX, REST APIs, TCP/IP, UDP Multicast

Practices & Tools: Git, CI/CD (GitHub Actions), Automated Testing, Agile/Scrum, Code Review, AI Coding Assistants (Claude Code, GitHub Copilot), Nginx, Postman

EXPERIENCE

Google Summer of Code 2026

May 2026 – Aug 2026

Kdenlive @KDE Community | Open Source Contributor, C++ and Qt

Remote

- Selected for **GSoC 2026** among 1,200 global contributors to design and ship Kdenlive's **RGB curve, gradient, and speed ramp editors in C++ and Qt**, owning features end to end from design through code review to release.
- Shipped **20 contributions** across Kdenlive and upstream repositories, resolving an **11-year-old bug** with a gap-detection algorithm now running for **41,000 monthly installs**, reviewed and merged by core maintainers.

Undergraduate Researcher

Jul 2025 – Nov 2025

Indian Institute of Technology, Jodhpur | Supervisor: Dr. Pratik Mazumder

IIT Jodhpur

- Engineered an offline **Retrieval-Augmented Generation pipeline** in Python, integrating **FAISS vector search** with 4-bit quantized Phi-3 and Flan-T5-XL under strict context-window and memory constraints.
- Benchmarked **prompt engineering** and retrieval strategies across **8 ROUGE metrics (+71.77% over baseline)**, reaching **3 second query latency** on 300-page PDFs for **15 users** across 3 feedback-driven iterations.

PROJECTS

Photon | [GitHub](#) | C++20, Linux, UDP Multicast, Lock-Free Concurrency

Jun 2026 – Jul 2026

- Built a **NASDAQ ITCH 5.0 feed handler and matching engine**: zero-copy parsing of packed wire structs, busy-pollled **recvmsg** UDP multicast, and a **lock-free SPSC ring** with cache-line-separated indices, reaching **38ns median parse** and **452ns median socket-to-consumed latency**, measured via **rdtscp** on a full day of real market data.
- Sustained **89k messages/sec** across **826 concurrent price-time priority order books** using array-indexed price levels, intrusive FIFO queues, and a fixed memory pool with **zero hot-path heap allocation**.
- Verified with **27 GoogleTest cases** including a two-thread ring stress test, **ASan/UBSan** sanitizer presets, and clang-tidy CI on every push.

gbemu | [GitHub](#) | C++20, SDL2, Emulation

Jul 2026

- Hand-wrote a **Game Boy (DMG) emulator** from hardware documentation: full **LR35902 instruction set** with cycle timing and the HALT bug, PPU with sprite priorities, **four sound channels**, and MBC0/1/3 cartridge mappers with battery saves.
- Passes all **11 Blargg cpu_instrs** CPU test ROMs and the **dmg-acid2** PPU accuracy test; runs commercial games including Pokemon Red.

DaemonDoc (team of 2) | [GitHub](#) | [Live](#) | TypeScript, LLM APIs, Express, MongoDB, Redis

Jan 2026 – May 2026

- Built an **agentic LLM pipeline** on a **Redis (BullMQ)** job queue with webhook idempotency and loop guards, sustaining an **85% success rate** across **959 production runs** over 68 repositories for **50 users**.
- Owned the **GitHub integration and prompt-construction layers**, plus a Razorpay billing system secured with **HMAC-SHA256** webhook signature verification.

COURSEWORK

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|----------------------------------|-------------------------------|------------------------------|
| • Data Structures and Algorithms | • Database Management Systems | • Pattern Recognition |
| • Object-Oriented Programming | • Operating Systems | • Probability and Statistics |
| • Software Engineering | • Machine Learning | • Discrete Mathematics |

ACHIEVEMENTS

- Solved **450+** algorithmic problems on [LeetCode](#) in C++; Max Rating **1429** on Codeforces across rated contests.
- JEE Advanced 2024: AIR **3449** of 180,000+ candidates; JEE Main 2024: **99.65** percentile, AIR **5680**.
- Merged **85+ open-source contributions** across GitHub and GitLab, shipping reviewed fixes to [Deno](#), [Adobe React Spectrum](#), [Kubernetes SIGs](#), [nlohmann/json](#), [Vitest](#), and [Kdenlive](#).

LEADERSHIP

- Junior Core Team Member**, DevUp Labs, IIT Jodhpur; mentored 30 students and ran 2 open-source workshops.
- Core Member**, Programming Club and E-Cell, IIT Jodhpur; co-organized 5 competitive programming contests, 10 sessions, and 3 entrepreneurship events.